# Valuing a Palantir (PLTR) Call Option:

## Volatility Evidence, Independent Pricing Engines and Asian-Option Variance Reduction

Underlying: Palantir Technologies Inc. (NASDAQ: PLTR) · Valuation date: 9 October 2026 (close)
Benchmark contract: PLTR 15 January 2027 call, strike $210

## Executive summary

We study one optionable, non-dividend-paying equity, Palantir (PLTR), on 9 October 2026. The benchmark is the 15 January 2027 \$210 call (98 days), quoted \$22.00/\$22.35. Its mid-quote implied volatility, $\sigma$ = 49.80%, is the common constant-volatility input. The main findings are:

- **Historical vs implied volatility (Q1).** Calibrated on 1,007 returns, PLTR's historical volatility is 67.3% (constant GBM). GARCH(1,1) has the lowest validation loss, but only 0.22% below constant volatility. All physical-measure forecasts exceed the risk-neutral ATM implied variance at every expiry, so the latest GARCH volatility is not the pricing volatility: it would price the call +43 half-spreads away from the market.
- **One contract, independent engines (Q2).** Monte Carlo, CRR and Jarrow–Rudd trees, and explicit, implicit and Crank–Nicolson finite differences all reproduce the Black–Scholes value (\$22.1750) within 0.15 market half-spreads. A transparent risk-neutral GARCH scenario prices the call at \$22.618 (control-variate SE 0.0081), +2.53 half-spreads from the mid. Almost all of this gap is a term-structure assumption about the initial variance, not numerical error.
- **Asian options (Q3).** Drawing each monitoring date from its marginal distribution independently, instead of building a Brownian path, cuts the ATM Asian value from \$13.41 to \$2.87. Under an equal budget of $2^{20}$ normal draws, geometric-average control variates improve efficiency by two to three orders of magnitude. We recommend the antithetic + pilot-$\beta$ control variate for routine use.
- **Reproducibility.** Every reported number was regenerated from the archived snapshot by one script. Fixed seeds, software versions, analytic benchmarks and numerical diagnostics are documented in Appendix B.

## 1. Underlying, valuation snapshot and conventions

**Table 1:** Valuation snapshot and conventions.

| Item | Value |
|---|---|
| Underlying | Palantir Technologies Inc. (PLTR); no cash dividend has been paid or announced, so none falls between valuation and any expiry used |
| Valuation date / time | 9 Oct 2026, market close 16:00 ET (20:00 UTC); option quotes are the 9 Oct end-of-day quotes (latest last-trade time 19:59:58 UTC = 15:59:58 ET) |
| Snapshot collection | Yahoo Finance via yfinance; chain downloaded 10 Oct 2026 10:37:56 UTC (18:37 SGT), after the close, so stock and option prices refer to the same 9 Oct close |
| Spot $S_0$ | \$209.05 (9 Oct close). Check: put–call parity on matched mid-quotes implies S between 208.09 and 209.27 (median 208.89) over 24 strike/expiry pairs |
| Risk-free rate r | 4.5% p.a., continuously compounded (non-negative) |
| Option chain | 4 expiries (2026-11-20 (42 d), 2026-12-18 (70 d), 2027-01-15 (98 d), 2027-02-19 (133 d)); 246 two-sided call quotes; $\geq$ 12 strikes on each side of $S_0$ at every expiry; matched puts near the money |
| Price history | 1,260 daily prices 2021-10-01 – 2026-10-08 (> 4 years) |
| Day counts | Returns: 252 trading days per year ($\Delta t = 1/252$). Option maturities: calendar days/365 (ACT/365), the market convention for option expiry dates and the basis of quoted implied volatilities; the GARCH scenario maps 68 trading-day steps onto the same T (Section 3.5) |

*The archived metadata file also contains spot_reference = 198.72 dated 8 Oct 2026. That is the previous day's close, and the option quotes are inconsistent with it: pricing at 198.72 would move the BS value by -30.9 half-spreads. All calculations therefore use the 9 Oct close of \$209.05.*

## 2. Question 1 — Historical and implied volatility as competing evidence

### 2.1 Data audit and validation design (Q1a)

We use the **adjusted close**, the series that is consistent with total returns. For PLTR it coincides exactly with the close (maximum absolute difference 0.0), because the company paid no dividends and had no splits in the sample: no daily price ratio is consistent with a split. The largest absolute daily log return, 26.9% on 2024-02-06, is an earnings-announcement move, not a corporate action. The audit found 0 missing values, 0 duplicate dates, 0 non-positive prices and 0 high/low inconsistencies. The daily log return is $r_t = \ln(P_t/P_{t-1})$ with $\Delta t = 1/252$, so annualised variance is 252 $\times$ daily variance. The 1,259 returns are split chronologically:

- calibration: the first 1,007 returns (2021-10-04 – 2025-10-07), the only data used to select $\lambda$ and estimate GARCH;
- validation: the most recent 252 returns (20%; 2025-10-08 – 2026-10-08), untouched until the forecasts are evaluated.

### 2.2 Three physical-measure models (Q1b)

**Constant-volatility GBM.** $\hat{\sigma} = \mathrm{sd}(r_t)\cdot\sqrt{252} = 67.29\%$. **EWMA.** $h_{t+1} = \lambda h_t + (1-\lambda) r_t^2$, started from the calibration sample variance. $\lambda$ is chosen by the in-sample MSE of $h_t$ against $r_t^2$, excluding a 30-day burn-in. The selected value, $\lambda$ **= 0.97**, has the lowest loss (Table 2). **GARCH(1,1).** $h_{t+1} = c + a(r_t - \mu)^2 + b\, h_t$, fitted by Gaussian maximum likelihood (arch package) on percentage returns.

All GARCH restrictions hold. **Persistence** a + b = 0.806 is the share of today's variance deviation from its long-run level that survives to tomorrow. It implies a half-life of only 3.2 trading days, short for an equity. **Volatility clustering** enters through a: a large squared return raises the next day's variance, so large moves tend to follow large moves. The long-run variance implies 67.4% p.a., almost identical to the constant-GBM estimate. The robust standard errors are large, and a, b

**Table 2:** EWMA $\lambda$ selection (calibration sample) and GARCH(1,1) estimates (robust standard errors).

| | Calib. MSE / estimate | Std. error | Note |
|---|---|---|---|
| EWMA $\lambda = 0.90$ | $2.5722\times10^{-5}$ | — | |
| EWMA $\lambda = 0.94$ | $2.5308\times10^{-5}$ | — | |
| EWMA $\lambda = 0.97$ | $2.5026\times10^{-5}$ | — | selected |
| GARCH $\mu$ (%/day) | 0.1962 | 0.1376 | |
| GARCH c = $\omega$ ($\%^2$) | 3.5112 | 18.800 | c > 0 ✓ |
| GARCH a = $\alpha$ | 0.0487 | 0.2092 | a $\geq$ 0 ✓ |
| GARCH b = $\beta$ | 0.7568 | 1.2383 | b $\geq$ 0 ✓ |
| Persistence a + b | 0.8055 | — | a + b < 1 ✓ (covariance-stationary) |
| Unconditional variance c/(1−a−b) | 18.05 $\%^2$/day | — | = 67.45% p.a. |
| Half-life of a variance shock | 3.2 days | — | EWMA: 22.8 days |

and c are not individually significant: the likelihood is flat along the a–b ridge, so the economic reading is indicative rather than precise.

**Out-of-sample evaluation.** All three models produce one-step-ahead variance forecasts over the 252 validation days with their parameters frozen. The common loss is MSE($\hat{h}_t - r_t^2$).

**Table 3:** Validation loss (252 one-step-ahead forecasts, daily decimal variance).

| Model | MSE | Relative to constant GBM | Rank |
|---|---|---|---|
| Constant GBM | $2.0572\times10^{-5}$ | 1.0000 | 2 |
| EWMA (lambda=0.97) | $2.0760\times10^{-5}$ | 1.0091 | 3 |
| GARCH(1,1) | $2.0526\times10^{-5}$ | 0.9978 | 1 |

GARCH ranks first, but its loss is only 0.22% below the constant model. Squared returns are a **noisy proxy** for latent variance: $r_t^2 = h_t\, z_t^2$ with $E[z^2] = 1$ but $Var(z^2) = 2$ under normality, so even a perfect forecast leaves a large error. In the validation sample the standard deviation of $r_t^2$ is 3.3 times its mean, and the five largest days contribute 31% of $\Sigma r_t^2$. A ranking decided by differences below 1% is therefore not robust.

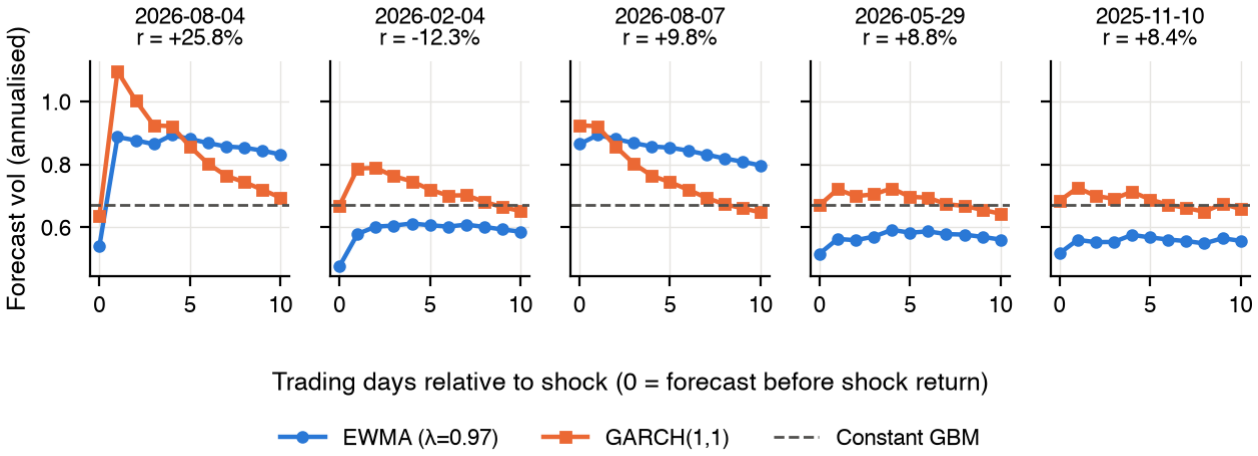



**Figure 1:** Annualised one-step forecast volatility around the five largest absolute validation returns (day 0 = forecast made before the shock return; days 1–10 = forecasts after it).

**Shock response, explained by the recursions.** The impact of a squared shock is $(1-\lambda)r^2 = 0.030r^2$ for EWMA and $a(r-\mu)^2 = 0.0487(r-\mu)^2$ for GARCH, so GARCH jumps further on impact. On 2026-08-04 (r = 25.8%), GARCH moves from 63.7% to 109.6%, while EWMA moves from 54.1% to 88.7%. Decay runs the other way. EWMA keeps $\lambda = 0.97$ of its variance each day (half-life 23 days) and has no long-run anchor, whereas GARCH reverts towards 67.4% at rate a + b = 0.806 (half-life 3.2 days). Ten days after the shock, GARCH is back near its long-run level while EWMA is still elevated (Figure 1). The same constant term explains why GARCH sits above EWMA before the smaller shocks: in calm periods EWMA drifts down towards recent squared returns, while GARCH is pulled up towards 67%. Both recursions respond only to $r^2$, so the negative 2026-02-04 shock is treated like a positive one. The only asymmetry is GARCH's $(r-\mu)$, with $\mu$ = 0.20%/day.

### 2.3 Implied-volatility structure (Q1c)

**Solver.** We use our own Black–Scholes call pricer and invert it with Brent's method on $\sigma \in [0.001, 5]$ (tolerance $10^{-10}$, at most 100 iterations). The root is bracketed only if the mid-quote lies strictly inside the no-arbitrage bounds $\max(S_0 - Ke^{-rT}, 0) < C < S_0$; otherwise no admissible volatility exists, and the quote is **flagged and retained, never deleted**. European inversion of the call is legitimate because, without dividends, early exercise of an American call is never optimal. Only mid-quotes are used. Of 247 call contracts, 246 have valid two-sided quotes, 226 invert successfully, and 20 deep in-the-money quotes lie below intrinsic value (wide spreads and stale quotes). These are plotted on a separate strip in Figure 2.

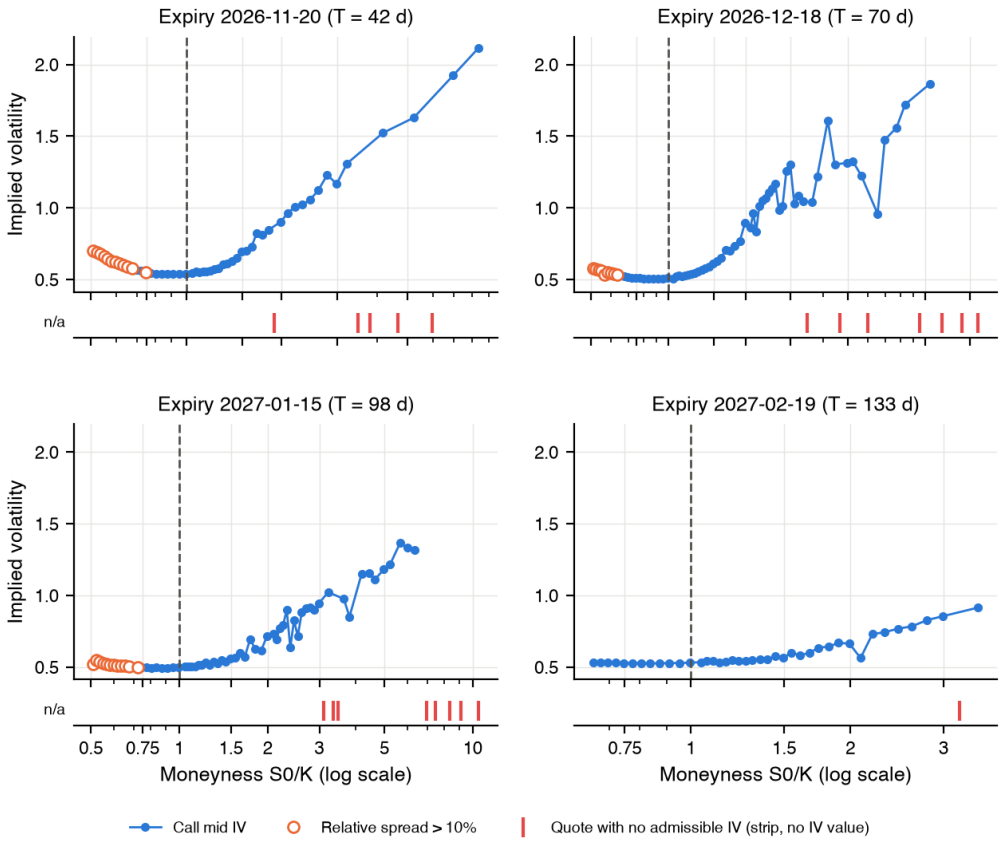



**Figure 2:** Call implied-volatility smiles for all four expiries against moneyness $S_0/K$. Open markers: relative spread above 10%. Bottom strips: quotes with no admissible implied volatility (20 mid-quotes below the no-arbitrage lower bound, plus one February quote with a zero bid).

The smile is flat near the money (about 50–54%) and rises steeply for in-the-money calls (high $S_0/K$). This is the usual equity skew, because a high-$S_0/K$ call corresponds by parity to an out-of-the-money put. Deep in-the-money implied volatilities are noisy: their wide spreads make the inversion highly sensitive to the mid.

**Table 4:** ATM (K = 210) implied volatility, total variance $w = \sigma^2 T$ and forward variance $(w_2 - w_1)/(T_2 - T_1)$.

| Expiry | Days | Mid | ATM IV | $w = \sigma^2 T$ | Forward interval | Fwd variance | Fwd vol |
|---|---|---|---|---|---|---|---|
| 2026-11-20 | 42 | 15.325 | 54.03% | 0.03359 | — | — | — |
| 2026-12-18 | 70 | 18.950 | 50.92% | 0.04972 | 2026-11-20 → 2026-12-18 | 0.2103 | 45.9% |
| 2027-01-15 | 98 | 22.175 | 49.80% | 0.06658 | 2026-12-18 → 2027-01-15 | 0.2199 | 46.9% |
| 2027-02-19 | 133 | 27.700 | 53.08% | 0.10265 | 2027-01-15 → 2027-02-19 | 0.3761 | 61.3% |

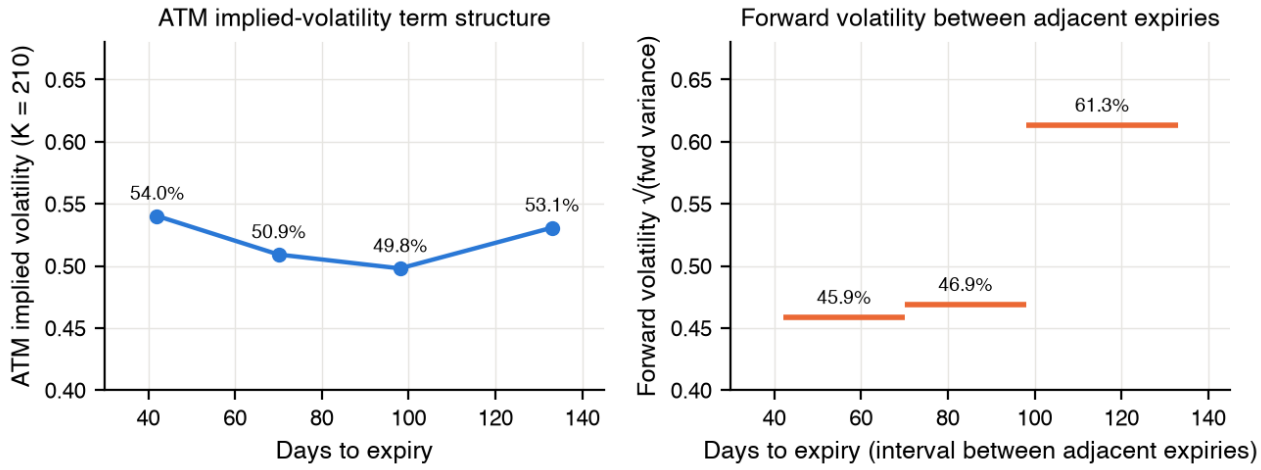



**Figure 3:** ATM implied-volatility term structure (left) and forward volatility between adjacent expiries (right).

The ATM term structure is U-shaped, but total variance increases monotonically. All 3 forward variances are positive, so there is no calendar-spread inconsistency at the money; none had to be investigated or replaced. The jump to a 61% forward volatility between January and February is consistent with an earnings announcement expected in that window.

The **P versus Q** comparison mixes two different objects. Implied variance comes from option prices under the risk-neutral measure. It embeds the market's expectation of future variance, any variance risk premium, and the Black–Scholes model

**Table 5:** Total variance to each expiry: risk-neutral (Q) ATM implied vs physical-measure (P) forecasts available at the 9 Oct close. P forecasts include the 9 Oct return; values using 8 Oct information are shown in brackets.

| Expiry | Q: $\sigma^2_{IV}$ T | P: GBM | P: EWMA | P: GARCH | P/Q (GARCH) |
|---|---|---|---|---|---|
| 2026-11-20 | 0.0336 | 0.0521 | 0.0379 (0.0368) | 0.0512 (0.0507) | 1.53 |
| 2026-12-18 | 0.0497 | 0.0868 | 0.0631 (0.0613) | 0.0855 (0.0850) | 1.72 |
| 2027-01-15 | 0.0666 | 0.1216 | 0.0884 (0.0858) | 0.1216 (0.1211) | 1.83 |
| 2027-02-19 | 0.1026 | 0.1650 | 0.1200 (0.1165) | 0.1650 (0.1644) | 1.61 |

*GARCH horizon variance: $\Sigma_{k=1..N}[V_L + (a+b)^{k-1}(h_{next} - V_L)]$, N = round(252T). EWMA has no mean reversion, so its forecast is flat: $h_{next} \times 252T$.*

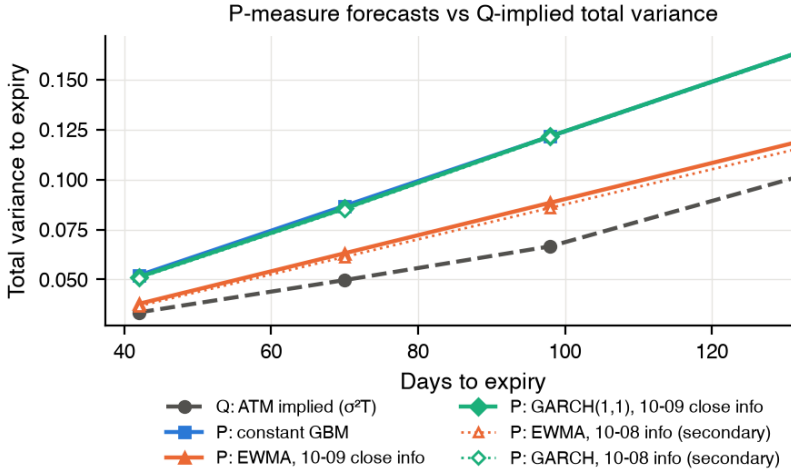



**Figure 4:** Physical-measure forecasts (P) against risk-neutral implied total variance (Q) at each option horizon. The constant-GBM line lies almost exactly under the GARCH line (e.g. 0.1216 vs 0.1216 at 98 days).

error. The historical forecasts are statistical predictions of realised variance under P. Every P forecast exceeds the Q value, by a factor of 1.13–1.83. GBM and GARCH nearly coincide because GARCH reverts to its 67% level within days. EWMA is closest to Q because it weights the recent, calmer year most heavily. Implied volatility below historical volatility is not an arbitrage: the market prices a calmer future than the 2022–2025 sample, which contained exceptional moves.

### 2.4 Evidence judgement (Q1d)

**Statement:** "Because GARCH gives the best out-of-sample return forecast, its latest conditional volatility is the correct volatility for option pricing." **We reject it, with one qualification.**

- **The forecasting win is not material.** GARCH beats constant volatility by 0.22% in MSE, using a loss built on a very noisy proxy and parameters that are not individually significant.
- **Wrong measure.** GARCH is fitted under P. A price is an expectation under Q, and the change of measure can alter the variance dynamics, not only the drift. Forecast accuracy under P says nothing about the volatility risk premium.
- **It fails the market test.** The horizon-equivalent GARCH volatility of 67.3% gives a BS value of $29.62, +43 half-spreads from the mid. The implied 49.8% prices the contract exactly by construction.
- **Qualification.** GARCH remains the better tool for P-measure risk measurement, and its dynamics can be a building block for pricing once they are mapped to Q and anchored to option prices, as Section 3.5 does.

## 3. Question 2 — One contract priced by independent numerical engines

The listed expiries are 42, 70, 98, 133 days away. Of the two that fall between 45 and 120 days, 98 days (15 Jan 2027) is closest to 90. We take the strike nearest the money, **K = 210** ($S_0/K = 0.9955$). Call quote 22.00/22.35 (mid 22.175, half-spread 0.175); matched put 20.60/21.25. The common input is the full-precision mid implied volatility, $\sigma$ **= 0.4979875393**, which reprices the mid to within $2.5\times10^{-14}$; T = 98/365 = 0.268493; r = 0.045.

### 3.1 Analytic benchmark and replication logic (Q2a)

Our own implementation gives $\mathbf{C_{BS}}$ **= $22.175000**, $\Delta = N(d_1) = 0.562878$ and $\Gamma = 0.0073036$ ($d_1 = 0.1583$, $d_2 = -0.0998$; vega 42.68 per unit $\sigma$). Parity, $P = C - S_0 + Ke^{-rT}$, gives a European put of **$20.6030**. This lies inside the observed put spread 20.60–21.25, 0.322 below the put mid (-0.99 put half-spreads). A gap of this size and sign is expected: the listed put is American, and with $r > 0$ early exercise of a deep put can be optimal, so its price includes an early-exercise premium that the European parity value lacks.

**Why the physical drift does not enter.** A short call can be hedged with $\Delta$ shares and borrowing, and the hedged portfolio is riskless, so it must earn r. The resulting PDE contains r and $\sigma$ but not the expected return $\mu$. Equivalently, pricing is an expectation under Q, where the stock drifts at r. The calibration-sample mean log return of 50% p.a. is irrelevant to the price. **Why the European value applies to the listed call.** With no dividends and $r \geq 0$, $C \geq S - Ke^{-r(T-t)} \geq S - K$. The

call is always worth at least its exercise value, so early exercise is never optimal and the American call equals the European call.

### 3.2 Monte Carlo price and delta (Q2b)

Terminal prices are simulated exactly: $S_T = S_0 \exp[(r - \sigma^2/2)T + \sigma\sqrt{T}\,Z]$. The SE is the sample SD of discounted payoffs divided by $\sqrt{N}$, and the 95% CI is $\pm 1.96$ SE. Each N uses its own child seed of SeedSequence(20261009) (PCG64).

**Table 6:** Crude Monte Carlo call price ($\sigma$ = 0.497988; exact terminal sampling; BS = 22.1750).

| N paths | Price | SE | 95% CI | \|error\| | z vs BS | Runtime (s) | Seed (child) |
|---|---|---|---|---|---|---|---|
| $2^{12}$ = 4,096 | 22.4244 | 0.5908 | [21.266, 23.582] | 0.2494 | +0.42 | 0.0005 | 0 |
| $2^{14}$ = 16,384 | 21.7401 | 0.2864 | [21.179, 22.301] | 0.4349 | -1.52 | 0.0004 | 1 |
| $2^{16}$ = 65,536 | 22.0839 | 0.1455 | [21.799, 22.369] | 0.0911 | -0.63 | 0.0029 | 2 |
| $2^{18}$ = 262,144 | 22.1485 | 0.0727 | [22.006, 22.291] | 0.0265 | -0.36 | 0.0060 | 3 |

Every interval covers the benchmark. The SE falls exactly as $N^{-1/2}$, a factor of 2 per fourfold increase in N (Figure 5, right), while the absolute error is non-monotone, as sampling noise implies.

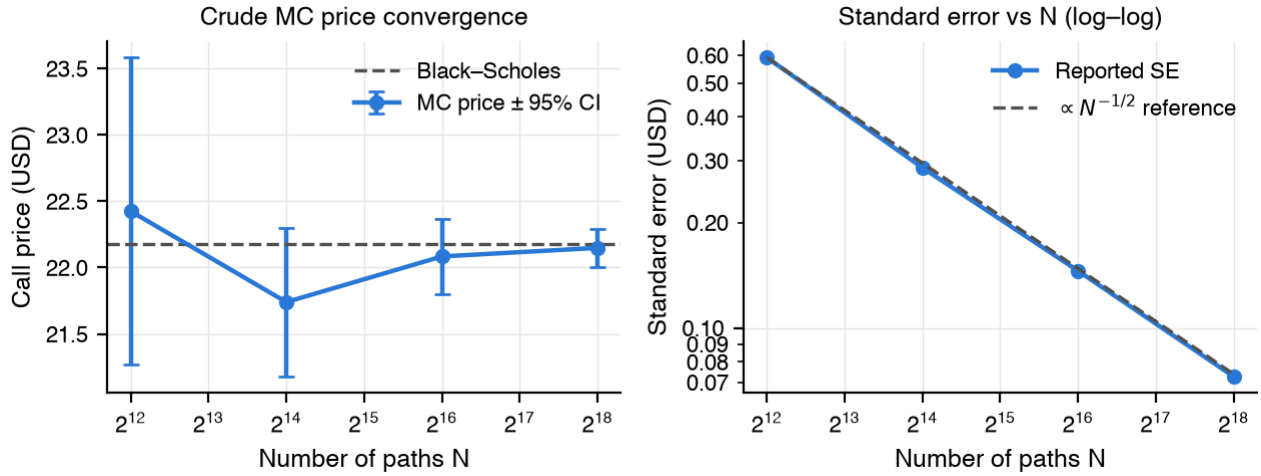



**Figure 5:** Crude MC price with 95% CI (left) and standard error against N on log–log axes (right).

**Table 7:** Monte Carlo delta, N = $2^{18}$ common paths (child seed 4). Analytic $\Delta$ = 0.562878.

| Estimator | $\Delta$ estimate | SE | Exact FD bias | Total error | Runtime (s) |
|---|---|---|---|---|---|
| CRN central difference, h = 0.5% | 0.562685 | 0.00122 | -0.000010 | -0.000194 | 0.0020 |
| CRN central difference, h = 1.0% | 0.562641 | 0.00121 | -0.000041 | -0.000237 | 0.0016 |
| CRN central difference, h = 2.0% | 0.562573 | 0.00120 | -0.000164 | -0.000306 | 0.0013 |
| Pathwise $e^{-rT}\cdot\mathbf{1}(S_T > K)\cdot S_T/S_0$ | 0.562859 | 0.00122 | +0.000000 | -0.000019 | 0.0011 |

*Exact FD bias = central difference applied to closed-form BS prices, i.e. the bump bias without sampling noise.*

All four estimators are statistically indistinguishable from the analytic delta ($|z| < 0.2$). With common random numbers (CRN), the central-difference bias is $O(h^2)$ ($-1.0\times10^{-5}$ at 0.5% rising to $-1.6\times10^{-4}$ at 2%), far below the SE of $\approx$0.0012. The pathwise estimator is unbiased and costs nothing extra. **A smaller bump need not be better** (Figure 6, right):

- Without CRN the two bumped prices are independent, so the SE grows like 1/h: at N = $2^{16}$ it is 0.049 at h = 1% and 4.91 at h = 0.01%.
- With CRN the call payoff is Lipschitz, so the SE stays flat ($\approx$0.0024), and shrinking h only trades a negligible bias for floating-point cancellation.
- For a discontinuous payoff, a digital call, even CRN fails: only paths ending within h of the strike contribute, so the SE rises from 0.00011 (h = 2%) to 0.00149 (h = 0.01%), roughly as $h^{-1/2}$.

### 3.3 Two binomial constructions (Q2c)

CRR uses $u = e^{\sigma\sqrt{\Delta t}}$, d = 1/u and $p = (e^{r\Delta t} - d)/(u - d)$. Jarrow–Rudd (JR) uses $p = \frac{1}{2}$ and puts the drift into the factors, u, $d = \exp[(r - \sigma^2/2)\Delta t \pm \sigma\sqrt{\Delta t}]$. Delta and gamma come from the time-$\Delta t$ and time-$2\Delta t$ nodes. Admissibility: CRR p lies in [0.4918, 0.4985]. For JR, the probability that would make the discounted stock a martingale on the lattice, $q = (e^{r\Delta t} - d)/(u - d)$, differs from $\frac{1}{2}$ by at most 5.7e-06. All probabilities are admissible.

**Diagnosis.** Neither tree converges monotonically. CRR's error changes sign between N = 25 and 50 and again between N = 200 and 400; JR's changes sign at N = 400. The cause is where the strike falls inside the terminal node grid. The error behaves like $c(\phi_N)/N$, where $\phi_N$ is the fractional position of $\ln(K/S_0)$ within a node cell. The right panel shows N $\times$ error collapsing onto one parabola in $\phi$ for both trees. Because CRR centres its lattice on $S_0$ and JR centres it on the risk-neutral

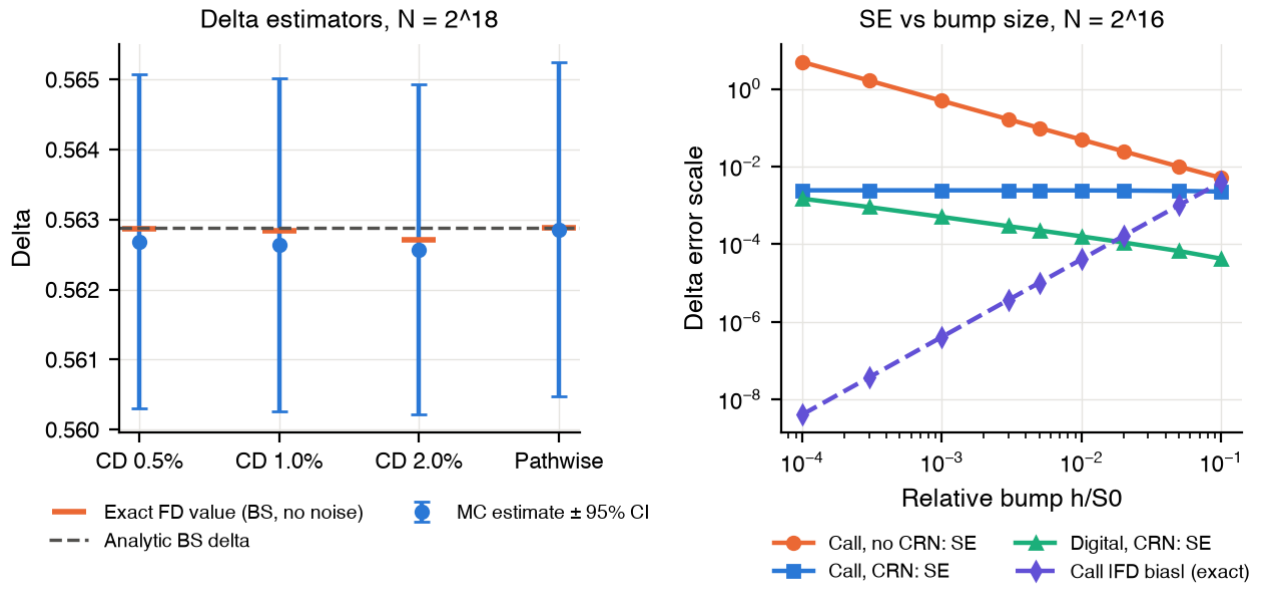



**Figure 6:** Delta estimators with 95% CI (left); estimator error scale against relative bump size (right; N = $2^{16}$, child seed 5).

**Table 8:** CRR and Jarrow–Rudd trees ($\sigma = 0.497988$). Runtime is the median of 5 runs, in ms. BS: $\Delta = 0.56288$, $\Gamma = 0.007304$.

| N | CRR price | error | Δ | Γ | ms | JR price | error | Δ | Γ | ms |
|---|---|---|---|---|---|---|---|---|---|---|
| 25 | 22.383265 | +0.208265 | 0.56275 | 0.007372 | 0.07 | 22.272296 | +0.097296 | 0.56135 | 0.007433 | 0.06 |
| 50 | 22.117581 | -0.057419 | 0.56264 | 0.007396 | 0.12 | 22.258675 | +0.083675 | 0.56248 | 0.007354 | 0.11 |
| 100 | 22.155550 | -0.019450 | 0.56277 | 0.007346 | 0.22 | 22.225949 | +0.050949 | 0.56265 | 0.007326 | 0.21 |
| 200 | 22.171361 | -0.003639 | 0.56284 | 0.007322 | 0.45 | 22.191452 | +0.016452 | 0.56273 | 0.007318 | 0.44 |
| 400 | 22.177006 | +0.002006 | 0.56286 | 0.007312 | 0.90 | 22.161406 | -0.013594 | 0.56277 | 0.007318 | 0.93 |
| 800 | 22.178229 | +0.003229 | 0.56287 | 0.007307 | 2.05 | 22.180942 | +0.005942 | 0.56285 | 0.007306 | 2.03 |

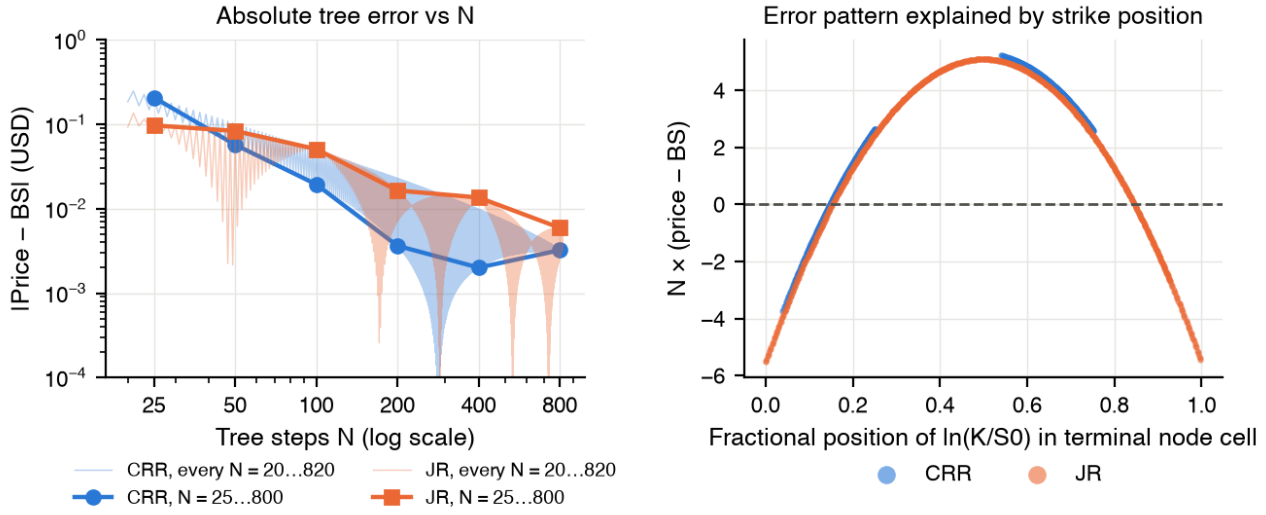



**Figure 7:** Tree error against N, with every N from 20 to 820 shown faintly (left), and N × error against the fractional position of $\ln(K/S_0)$ inside the terminal node cell (right).

drift, the two trees place the strike at different $\phi$ for the same N. They therefore disagree at finite N (at N = 800: CRR +0.0032, JR +0.0059), even though both match the first two moments of the same GBM increment and converge at O(1/N) to the same Black–Scholes value. Runtime grows as $O(N^2)$; both take about 1 ms at N = 800.

### 3.4 Finite-difference stability and convergence (Q2d)

**Transformation (Week 4).** With $x = \ln(S/K)$, $\tau = \sigma^2(T - t)/2$, $k = 2r/\sigma^2$, $a = (1 - k)/2$, $b = -(k + 1)^2/4$ and $C = K\, e^{ax + b\tau}\, u(x, \tau)$, the Black–Scholes PDE becomes $u_\tau = u_{xx}$ (here k = 0.3629, a = 0.3185, b = -0.4644, $\tau_{max} = 0.033292$). Each condition used in the code follows from mapping a payoff or asymptote through the same transformation:

- **Initial condition** (the payoff at $\tau = 0$): $(S - K)^+ = K(e^x - 1)^+ \Rightarrow u(x, 0) = e^{-ax}(e^x - 1)^+$.
- **Left boundary** ($S \to 0$, where $C \to 0$): $u(x_{min}, \tau) = 0$.
- **Right boundary** (S large, where $C \approx S - Ke^{-r(T-t)}$ and $r(T-t) = k\tau$): $u(x_{max}, \tau) = \exp(-a \cdot x_{max} - b\tau) \cdot (\exp(x_{max}) - e^{-k\tau})$.
- **Mapping back.** $C_j = K \cdot \exp(a \cdot x_j + b \cdot \tau_{max}) \cdot u_j$ on the grid, with $C(S_0)$ interpolated at $x_0 = \ln(S_0/K) = -0.00453$. Delta

and gamma come from finite differences of $C_j$ in S.

Implicit and Crank–Nicolson (CN) solve one tridiagonal system per time step. We factor the system once with our own LU (Thomas) decomposition and reuse the factors. The explicit scheme is stable only for $\alpha = \Delta\tau/\Delta x^2 \leq \frac{1}{2}$.

**Table 9:** Explicit scheme on x ∈ [−1, 1]: three stable refinements ($\alpha \approx 0.40$), then $\alpha > \frac{1}{2}$ on the same M = 100 grid.

| M | N | Δx | Δτ | α | Price | Error | Δ | Γ | Runtime (s) |
|---|---|---|---|---|---|---|---|---|---|
| 100 | 209 | 0.020 | $1.59\times10^{-4}$ | 0.398 | 22.182729 | +0.007729 | 0.56264 | 0.007285 | 0.0010 |
| 200 | 833 | 0.010 | $4.00\times10^{-5}$ | 0.400 | 22.178099 | +0.003099 | 0.56281 | 0.007299 | 0.0042 |
| 400 | 3330 | 0.005 | $1.00\times10^{-5}$ | 0.400 | 22.175123 | +0.000123 | 0.56287 | 0.007302 | 0.0170 |
| 100 | 139 | 0.020 | $2.40\times10^{-4}$ | 0.599 | $2.60\times10^{18}$ | unstable | — | — | — |
| 100 | 105 | 0.020 | $3.17\times10^{-4}$ | 0.793 | $5.66\times10^{33}$ | unstable | — | — | — |
| 100 | 84 | 0.020 | $3.96\times10^{-4}$ | 0.991 | $-1.28\times10^{38}$ | unstable | — | — | — |

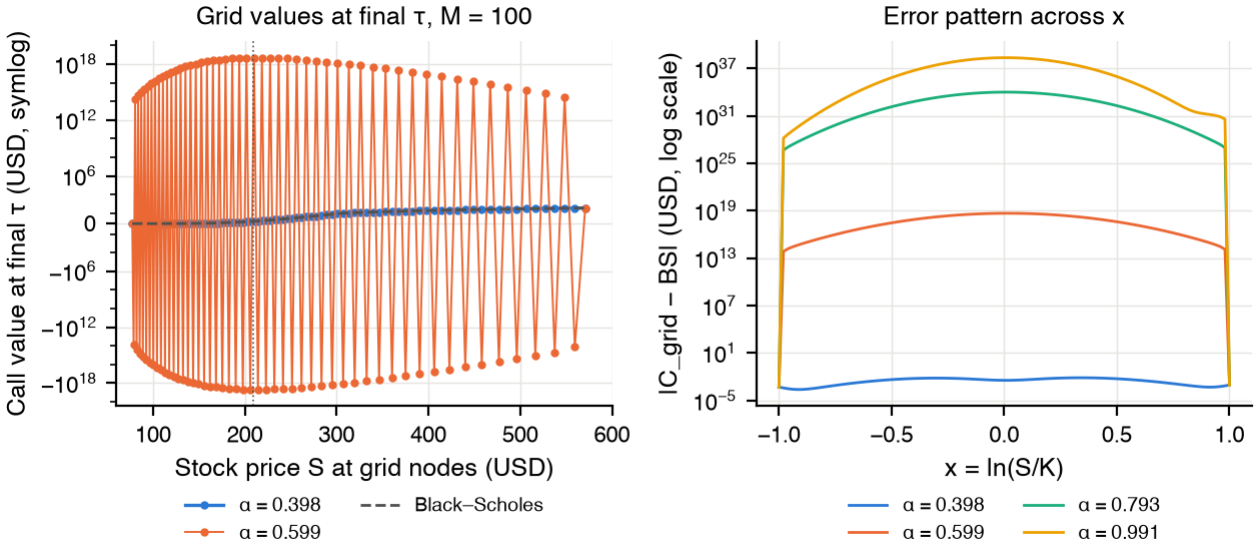



**Figure 8:** Explicit scheme on a fixed M = 100 grid: grid values at $\tau_{max}$ (left, symmetric-log scale) and |error| across the grid (right). Once $\alpha > \frac{1}{2}$ the highest mode grows by $|1 - 4\alpha|$ per step, producing a sawtooth over the whole grid.

The instability shows in the grid values, not just the price. With $\alpha = 0.60$, the highest-frequency Fourier mode is multiplied by $|1 - 4\alpha| = 1.40$ per step. The result is an alternating sawtooth reaching $10^{18}$ in every interior node. The truncated domain and boundary values are unchanged; only $\Delta\tau$ differs.

**Table 10:** Explicit, fully implicit and CN on identical grids (x ∈ [−1, 1], $\alpha = 0.416$).

| Scheme | M | N | Price | Error | Δ | Γ | Runtime (s) |
|---|---|---|---|---|---|---|---|
| Explicit | 100 | 200 | 22.183285 | +0.008285 | 0.562644 | 0.0072842 | 0.0010 |
| Fully implicit | 100 | 200 | 22.157505 | -0.017495 | 0.562534 | 0.0073114 | 0.0113 |
| Crank-Nicolson | 100 | 200 | 22.170408 | -0.004592 | 0.562589 | 0.0072978 | 0.0112 |
| Explicit | 200 | 800 | 22.178226 | +0.003226 | 0.562810 | 0.0072985 | 0.0040 |
| Fully implicit | 200 | 800 | 22.171786 | -0.003214 | 0.562783 | 0.0073053 | 0.0867 |
| Crank-Nicolson | 200 | 800 | 22.175007 | +0.000007 | 0.562797 | 0.0073019 | 0.0868 |
| Explicit | 400 | 3200 | 22.175155 | +0.000155 | 0.562866 | 0.0073025 | 0.0158 |
| Fully implicit | 400 | 3200 | 22.173545 | -0.001455 | 0.562860 | 0.0073042 | 0.7050 |
| Crank-Nicolson | 400 | 3200 | 22.174350 | -0.000650 | 0.562863 | 0.0073033 | 0.7073 |

**Separating the error sources** (Figure 9). (i) Domain truncation: at fixed $\Delta x = 0.01$ and N = 800, widening the domain from ±1 to ±1.5 changes CN by $2.5\times10^{-10}$, while ±0.5 (S ∈ [127, 346]) costs -0.0013. At fixed M = 200 the apparent range effect is dominated by the change in Δx, which is why Δx must be held fixed. (ii) Time: at M = 200, doubling N halves the implicit change (first order in $\Delta\tau$) and quarters the CN change (second order). (iii) Space: at N = 6400, CN's error is non-monotone in M (+0.000006 at M = 200, -0.000650 at M = 400). Linear interpolation in S has bias $\approx \Gamma/2 \cdot (S_0 - S_i)(S_{i+1} - S_0)$, which depends on where $S_0$ sits in its cell; it cancels against the discretisation error by chance at M = 200, so the small error there is luck rather than accuracy. By M = 1600 the error is -0.000003.

**Recommended grid: Crank–Nicolson, x ∈ [−1, 1] (S ∈ [77, 571]), M = 800, N = 3200** ($\Delta x = 0.0025$, $\Delta\tau = 1.04\times10^{-5}$, $\alpha = 1.66$). It gives price 22.174904 (error -0.000096, i.e. -0.0005 half-spreads), $\Delta = 0.562874$ and $\Gamma = 0.0073035$, in 1.44 s, with a robust error bound of 3.4e-04. The evidence: truncation is below $10^{-9}$ at ±1; the time error is below $10^{-7}$ for N ≥ 3200 at M = 200; the price lies within $10^{-4}$ of the finest grid (M = 1600, N = 6400); and the grid gamma never deviates from BS by more than 1.9e-07 near the money, so there are no kink oscillations. The cheaper N = 400 gives the same price to 4e-06, with the same gamma accuracy, so no oscillation is visible there. We still choose N = 3200 as a safety margin: at $\alpha = 13.3$, CN damps the high-frequency mode excited by the payoff kink only weakly (per-step factor $\approx 0.93$, against 0.54 at $\alpha \approx 1.7$), and the extra cost is under 1 s. Fully implicit on the same grid is first-order in time (error -0.00090).

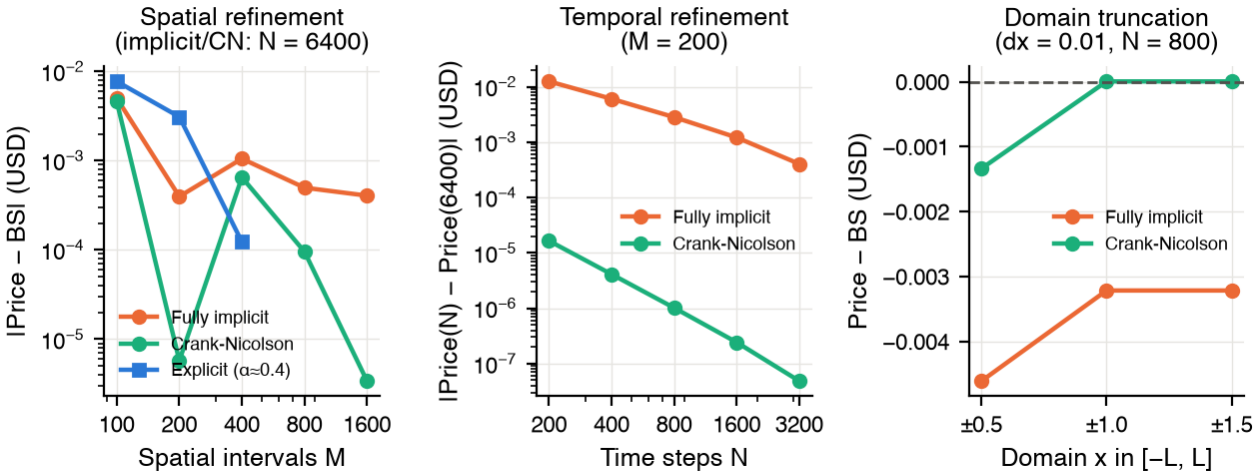



**Figure 9:** Spatial refinement at N = 6400 (left), temporal refinement at M = 200 (centre) and domain truncation at fixed $\Delta x$ (right).

### 3.5 A transparent risk-neutral GARCH scenario (Q2e)

The Q1 GARCH parameters describe P. The working Q-scenario makes four explicit choices:

- **Dynamics:** keep the fitted a = 0.0487 and b = 0.7568 (shock response and persistence), with 68 trading-day steps of length $\Delta$ = T/68.
- **Long-run level:** set the per-step long-run variance to $v^* = \sigma_{IV}^2 T/68$ (9.792 %$^2$), i.e. $\omega_Q = (1 - a - b)v^*$. A path that starts at $v^*$ then has the same expected total variance as Black–Scholes.
- **Initial state:** $h_0$ = 15.889 %$^2$ (63.3% p.a.), the P-filtered variance after the 9 Oct return (5.04%).
- **Risk-neutral drift:** $\ln S_{i+1} = \ln S_i + r\Delta - h_i/2 + \sqrt{h_i}\, z_i$, so $e^{-rt}S_t$ is a martingale (simulation check: z = -1.00).

To separate sampling noise from model effects, every run carries a paired control variate: a constant-$\sigma$ GBM driven by the same $z_i$, whose mean is known exactly (the BS price). Its coefficient, $\beta$ = 1.0233, is estimated on an independent 20,000-path pilot.

**Table 11:** GARCH-Q convergence, primary $h_0$. Crude MC and control-variate (CV) estimates; 68 steps; seeds = child 6–11 of SeedSequence(20261009).

| Paths | Crude price ± SE | CV price | CV SE | CV 95% CI | CV − BS (z) | Runtime (s) |
|---|---|---|---|---|---|---|
| 1,000 | 22.046 ± 1.157 | 22.6461 | 0.1127 | [22.425, 22.867] | +0.471 (4.2) | 0.001 |
| 5,000 | 22.893 ± 0.551 | 22.6218 | 0.0526 | [22.519, 22.725] | +0.447 (8.5) | 0.006 |
| 10,000 | 22.502 ± 0.383 | 22.6803 | 0.0372 | [22.607, 22.753] | +0.505 (13.6) | 0.010 |
| 50,000 | 22.734 ± 0.173 | 22.6151 | 0.0160 | [22.584, 22.646] | +0.440 (27.5) | 0.038 |
| 100,000 | 22.503 ± 0.121 | 22.6199 | 0.0114 | [22.598, 22.642] | +0.445 (39.0) | 0.079 |
| 200,000 | 22.520 ± 0.086 | 22.6179 | 0.0081 | [22.602, 22.634] | +0.443 (54.9) | 0.229 |

**Table 12:** Sensitivity to the initial variance $h_0$ (200,000 paths each, independent seeds).

| Initial state $h_0$ | $h_0$ (%$^2$) | Ann. vol | Q-expected vol to T | CV price ± SE | CV − BS | D (half-spr.) |
|---|---|---|---|---|---|---|
| calibration-end state 2025-10-07 [old code] | 17.598 | 66.6% | 51.28% | 22.7687 ± 0.0091 | +0.5937 | +3.39 |
| one-step forecast for 2026-10-09 | 14.848 | 61.2% | 50.76% | 22.5315 ± 0.0074 | +0.3565 | +2.04 |
| post-2026-10-09 state (primary) | 15.889 | 63.3% | 50.96% | 22.6356 ± 0.0080 | +0.4606 | +2.63 |
| target $v^*$ (= BS total variance) | 9.792 | 49.7% | 49.80% | 22.1326 ± 0.0057 | -0.0424 | -0.24 |

**Result.** The scenario value is **$22.6179** (95% CI [22.602, 22.634]), $0.443 or +2.53 half-spreads above the mid and outside the quoted spread. The control variate cuts the variance by a factor of 112, so the gap is not sampling error. The independent replicate in the $h_0$ table gives 22.6356; the two replicates differ by 0.018 (≈1.6 joint SE). Using the $h_0$-table runs throughout, the gap of +0.461 decomposes as follows:

- **Term-structure effect, +0.503.** Starting at 63% rather than the long-run 49.8%, variance decays over a few days, raising the Q-expected volatility to expiry to 50.96%.
- **Pure dynamics effect, -0.0424** (z = -7.5). Holding expected total variance equal to BS, GARCH's stochastic variance (fatter-tailed $S_T$) slightly lowers the value of this near-ATM call. The effect is statistically significant but economically small (-0.24 half-spreads).

**What is and is not calibrated.** Only one number is matched to the market: the long-run variance level, to a single ATM implied volatility. The parameters a and b, the initial state, the shape of the risk premium and the smile are all inherited from P or simply assumed. The old notebook used the variance at the end of the calibration sample (Oct 2025, 17.6 %$^2$) as $h_0$. Correcting it moves the price by -0.133, and $h_0$ choices alone span 0.64 (≈3.6 half-spreads). **Non-uniqueness and incompleteness.** With stochastic variance driven by the same shock, but not spanned by the stock and the bond in continuous trading, there are infinitely many equivalent martingale measures. Duan-type risk premia, a different long-run level or another $h_0$ are all arbitrage-free. Without more instruments (other options, variance swaps) no-arbitrage alone

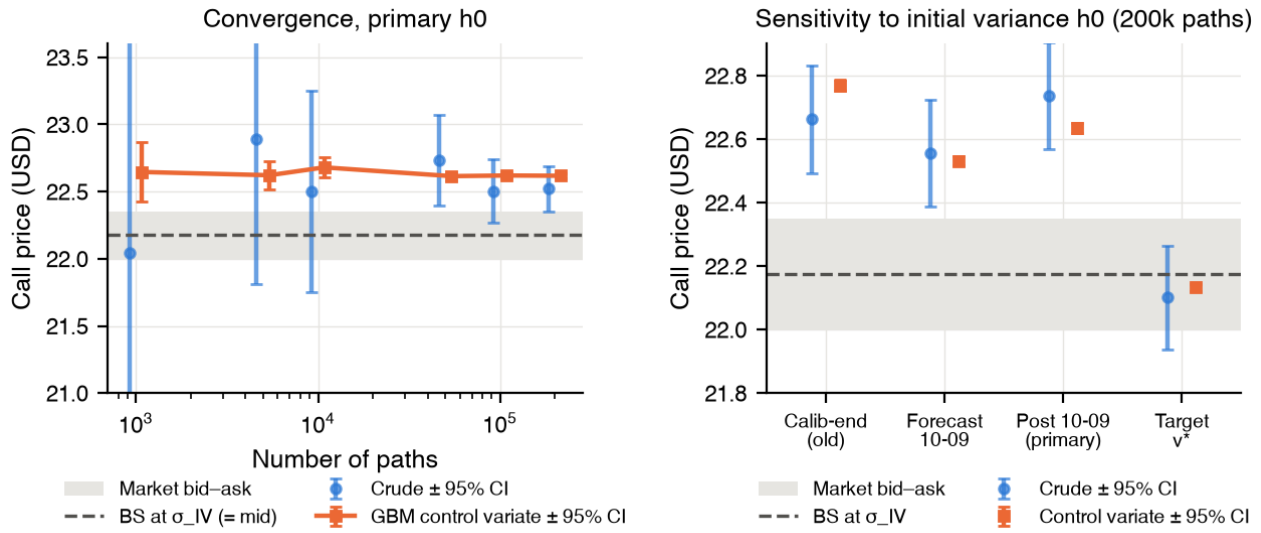



**Figure 10:** GARCH-Q scenario: convergence of crude and control-variate estimates (left; the crude 95% intervals at 1,000 and 5,000 paths are clipped by the axis, see Table 11); sensitivity to the initial variance (right). Grey band: market bid–ask.

cannot pick one, so the GARCH value is a scenario, not a price prediction. The market's ATM term structure (Table 4) implies a lower near-term volatility than the start state assumes.

### 3.6 Reconciliation and model risk (Q2f)

**Table 13:** Reconciliation (mid 22.175, half-spread 0.175). D = (price − mid)/half-spread; $|D| \leq 1$ lies inside the quoted spread. Deterministic error is measured against BS.

| Method | Volatility | Settings | Price | 95% CI / det. error | Runtime (s) | D |
|---|---|---|---|---|---|---|
| Black–Scholes | $\sigma_{IV}$ const. | — | 22.175000 | exact | — | +0.000 |
| Crude Monte Carlo | $\sigma_{IV}$ const. | $2^{18}$ paths | 22.148523 | [22.006, 22.291] | 0.0060 | -0.151 |
| CRR binomial tree | $\sigma_{IV}$ const. | N = 800 steps | 22.178229 | +0.003229 | 0.0021 | +0.018 |
| Jarrow–Rudd binomial tree | $\sigma_{IV}$ const. | N = 800 steps | 22.180942 | +0.005942 | 0.0020 | +0.034 |
| Explicit FDM | $\sigma_{IV}$ const. | x∈[-1,1], M=400, N=3330, dx=0.005, dτ=9.998e-06, α=0.400 | 22.175123 | +0.000123 | 0.0170 | +0.001 |
| Fully implicit FDM | $\sigma_{IV}$ const. | x∈[-1,1], M=800, N=3200, dx=0.0025, dτ=1.040e-05, α=1.665 | 22.174099 | -0.000901 | 1.4467 | -0.005 |
| Crank–Nicolson FDM | $\sigma_{IV}$ const. | x∈[-1,1], M=800, N=3200, dx=0.0025, dτ=1.040e-05, α=1.665 | 22.174904 | -0.000096 | 1.4356 | -0.001 |
| GARCH(1,1) Q-scenario MC | GARCH-Q (a, b from P; $v^* = \sigma_{IV}^2 T/68$; $h_0$ post-9-Oct) | 200k × 68 steps, GBM CV | 22.617856 | [22.602, 22.634] (SE 0.0081) | 0.2292 | +2.531 |

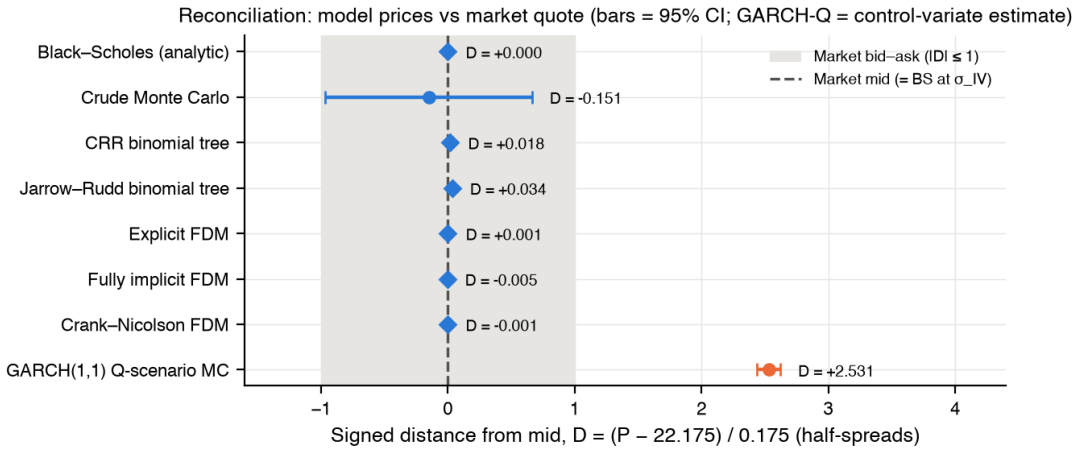



**Figure 11:** Signed distance of every engine from the mid-quote, in half-spreads, with 95% CIs.

**Classification of discrepancies**, each backed by a numerical check (Table 14; full list in results/tables/T10):

**Table 14:** Discrepancy classes and supporting checks.

| Class | Numerical evidence | Conclusion |
|---|---|---|
| Quote noise | Half-spread 0.175 = 0.79% of mid. Parity-implied spot ranges over 1.18 USD across strikes (≈3.8 call half-spreads via $\Delta$). The European parity put is -0.99 put half-spreads below the American put mid. | The market price is known only to $\pm 1$ half-spread. |
| Input uncertainty | $\pm 1$ vol-pt moves BS by $\pm 0.427$ ($\pm 2.44$ half-spreads). The bid/ask IVs (49.39%/50.21%) span $\pm 1$ half-spread. Using the stale 198.72 spot would shift the price by -31 half-spreads. | Dominates every numerical error by 2–4 orders of magnitude. |
| Sampling error | MC $2^{18}$ z = -0.36; GARCH-Q crude SE 0.086 vs CV SE 0.0081; martingale z = -1.00. | MC deviations are pure noise. |
| Discretisation error | CRR(800) − CRR(400) = +0.0012; JR +0.0195; CN recommended − finest = $-9.2\times10^{-5}$; implicit -0.0009 (first order). | Deterministic, $\leq 0.034$ half-spreads. |
| Model error | GARCH-Q − BS = +0.443 (z = 55); of which the $h_0$ term structure is +0.503 and the pure dynamics -0.0424 (z = -7.5). | The only material discrepancy: an assumption, not a bug. |

**Assessment of the statement** "If BS, MC, a binomial tree and CN agree to four decimal places, the market option has been correctly priced." **Reject.** The agreement shows only that four algorithms solve the same mathematical problem consistently: our constant-$\sigma$ engines agree within 0.04 half-spreads. All of them share the same inputs and assumptions. $\sigma$ is backed out of the very quote being 'tested', so the agreement is circular. The model ignores the smile, stochastic volatility and the American features of the puts. The quote itself is uncertain by $\pm 1$ half-spread, and a 1 vol-point change in the shared input moves every engine together by 2.4 half-spreads. The GARCH-Q scenario, equally arbitrage-free, differs by 2.5 half-spreads. Numerical agreement is necessary for a credible price, but it says nothing about model or input risk.

## 4. Question 3 — Asian option estimation and variance reduction

We use the Q2 inputs ($S_0$ = 209.05, r = 0.045, T = 0.268493, $\sigma$ = 0.497988), m = 52 monitoring dates $t_i = iT/52$ ($S_0$ excluded from the average) and strikes $0.9S_0$, $S_0$ and $1.1S_0$ ($188.145, $209.05 and $229.955). The payoff is $X = e^{-rT}(A - K)^+$ with arithmetic average $A = (1/m)\Sigma\, S(t_i)$.

### 4.1 Path construction and the dependence check (Q3a)

Each path is built from 52 independent N(0,1) increments with exact log transitions, $\ln S(t_i) = \ln S(t_{i-1}) + (r - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}\, Z_i$. The **dimension is 52**: one path uses 52 scalar normals, and the $2^{20}$ budget buys $\lfloor 2^{20}/52 \rfloor$ = 20,164 paths (1,048,528 draws). The **incorrect alternative** draws each $S(t_i)$ independently from its correct marginal: $\ln S(t_i) = \ln S_0 + (r - \sigma^2/2)t_i + \sigma\sqrt{t_i}\, Z_i$, with independent $Z_i$ at each date.

**Table 15:** Correct path vs independent marginals (20,164 paths; 1,048,528 normals each; seed SeedSequence([20261019, 1000]); runtime 0.028 s for both).

| Strike | Correct path: price (SE) | Independent marginals: price (SE) | Difference (paired SE) |
|---|---|---|---|
| $0.9S_0$ | 26.1863 (0.1918) | 22.0048 (0.0378) | 4.1815 (0.1706) |
| $S_0$ (ATM) | 13.4069 (0.1491) | 2.8714 (0.0257) | 10.5355 (0.1352) |
| $1.1S_0$ | 5.8695 (0.1026) | 0.0004 (0.0001) | 5.8691 (0.1026) |

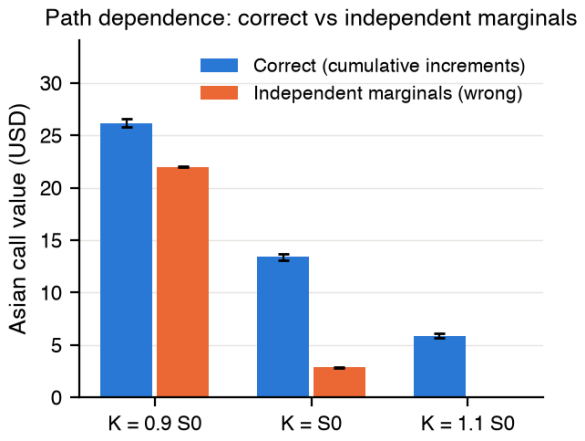



**Figure 12:** Asian call values under correct and independent-marginal path construction ($\pm 1.96$ SE).

Each date's marginal law is right, but the joint law is wrong. Brownian motion has $\mathrm{Cov}(W_s, W_t) = \min(s, t)$, so the correct log prices have covariance $\sigma^2 \min(t_i, t_j)$, whereas the alternative sets every cross-date covariance to zero. For the log-average this gives $\mathrm{Var} = \sigma^2 T(m+1)(2m+1)/(6m^2) = \sigma^2 \cdot 0.0921$ (correct) against $\sigma^2 T(m+1)/(2m^2) = \sigma^2 \cdot 0.00263$ (independent), a ratio of (2m+1)/3 = 35. Independent noise averages out, so the effective annualised volatility of the log-average collapses from 29.2% to 4.9%, and with it the optionality: the ATM value falls from 13.41 to 2.87, and the $1.1S_0$ call becomes almost worthless. The price levels come from one crude run per strike (each within 2 SE of the high-precision values in Section 4.3), so the evidence is the paired difference: 10.54 at the money with SE 0.14, i.e. about 78 SE.

### 4.2 Five estimators (Q3b)

All five estimators share one path generator. The control is the discounted geometric-average call $Y = e^{-rT}(G - K)^+$ with $G = \exp[(1/m)\Sigma \ln S(t_i)]$. Since $\ln G \sim N(\ln S_0 + (r - \sigma^2/2)T(m+1)/(2m), \sigma^2 T(m+1)(2m+1)/(6m^2))$, its mean $\mu_Y$ has the closed form $\mu_Y = e^{-rT}[\exp(\mu_G + v_G/2)\cdot N(d_1) - KN(d_2)]$, which gives 24.9968, 12.5075 and 5.2800 for the three strikes. The estimators are:

- **(1) Crude:** $\bar{X}$.
- **(2) Antithetic:** pair-level means $\bar{X}_j = [X(Z_j) + X(-Z_j)]/2$; the SE uses the number of pairs.
- **(3) CV, $\beta = 1$:** $\bar{X} - (\bar{Y} - \mu_Y)$.
- **(4) CV, pilot $\beta$:** $\hat{\beta} = \mathrm{Cov}(X, Y)/\mathrm{Var}(Y)$ estimated on an independent pilot of 4,032 paths, then $\bar{X} - \hat{\beta}(\bar{Y} - \mu_Y)$ on the remaining 16,132. The pilot counts towards the budget.
- **(5) Antithetic + pilot CV:** the paired observations $(\bar{X}_j, \bar{Y}_j)$ are defined first; $\beta$ is estimated from pilot pairs and applied to production pairs.

### 4.3 Equal-cost efficiency study (Q3c)

**Table 16:** Budget per estimator run: $2^{20}$ = 1,048,576 scalar normals (1,048,528 used). Counts cover all three strikes.

| Estimator | Normal draws | Pilot / production paths | Stock paths | Target payoffs | Control payoffs | Total payoffs |
|---|---|---|---|---|---|---|
| Crude MC | 1,048,528 | 0 / 20,164 | 20,164 | 60,492 | 0 | 60,492 |
| Antithetic | 1,048,528 | 0 / 20,164 | 40,328 | 120,984 | 0 | 120,984 |
| CV beta=1 | 1,048,528 | 0 / 20,164 | 20,164 | 60,492 | 60,492 | 120,984 |
| CV pilot beta | 1,048,528 | 4,032 / 16,132 | 20,164 | 60,492 | 60,492 | 120,984 |
| Antithetic + pilot CV | 1,048,528 | 4,032 / 16,132 | 40,328 | 120,984 | 120,984 | 241,968 |

*Antithetic $-Z$ paths use no new normals but double the stock paths and payoff evaluations, which is why both counts are reported.*

**Table 17:** ATM Asian call, 30 independent macro-seeds per estimator (master seed 20261019, SeedSequence streams per repetition × estimator). RE = $(s^2 \cdot t)_{crude}/(s^2 \cdot t)_{method}$, with 95% bootstrap CIs (5,000 resamples).

| Estimator | Mean price [95% CI of mean] | Emp. SD | Avg. SE | SD/SE | Runtime (ms) | VRF | RE [95% CI] |
|---|---|---|---|---|---|---|---|
| Crude MC | 13.1215 [13.072, 13.171] | 0.1314 | 0.1468 | 0.89 | 18.2 | 1.0 | 1.00 [1.00, 1.00] |
| Antithetic | 13.1531 [13.126, 13.180] | 0.0712 | 0.0808 | 0.88 | 30.2 | 3.4 | 2.06 [0.93, 4.30] |
| CV beta=1 | 13.1286 [13.125, 13.132] | 0.0089 | 0.0079 | 1.13 | 19.8 | 217 | 200 [91, 406] |
| CV pilot beta | 13.1313 [13.129, 13.133] | 0.0057 | 0.0056 | 1.00 | 20.7 | 536 | 472 [210, 999] |
| Antithetic + pilot CV | 13.1312 [13.130, 13.133] | 0.0039 | 0.0039 | 1.00 | 33.0 | 1114 | 614 [284, 1266] |

*RE uses wall-clock time for all three strikes per run (Apple M5, Python 3.13.9/NumPy 2.5.4). With 30 repetitions an empirical SD has roughly ±13% relative error, so SD/SE ratios of 0.88–1.13 are consistent with correct standard errors.*

**Empirical SD vs reported SE.** For every estimator the run-to-run SD agrees with the average reported SE (ratios 0.88–1.13), so the standard errors can be trusted. The pilot-$\beta$ SE includes the first-order uncertainty of $\beta$. All estimators agree on the price, about 13.131. The control variates reduce variance 200–1,100-fold, while antithetic sampling alone gives only 3.4×, and its time-adjusted RE CI includes 1.

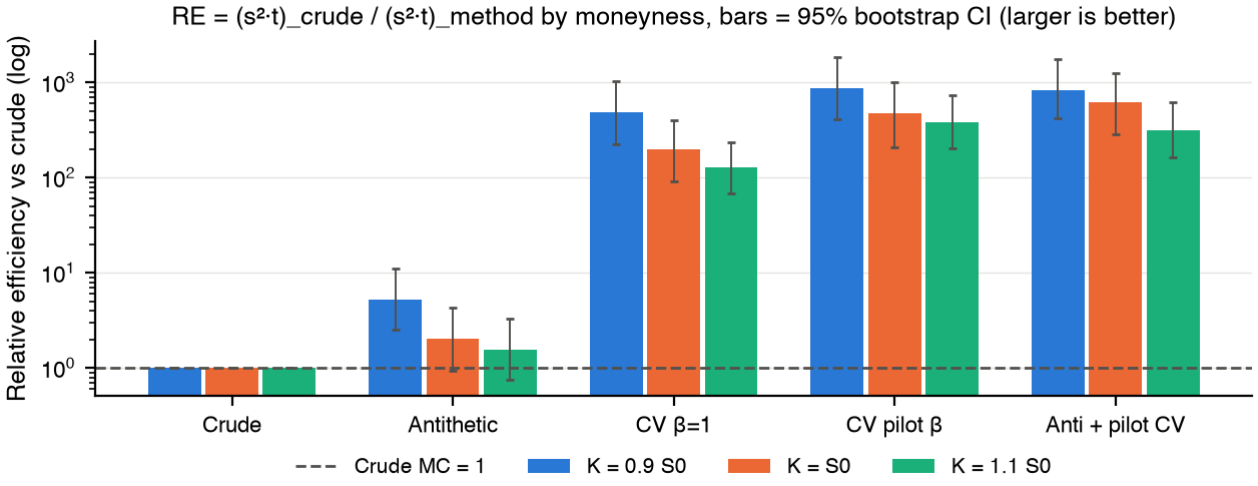



**Figure 13:** Relative efficiency by estimator and moneyness, with 95% bootstrap CIs (log scale).

**Why the ranking moves with moneyness.** (i) Correlation. The arithmetic–geometric payoff correlation falls from 0.99954 ($0.9S_0$) to 0.99896 ($1.1S_0$). The best achievable CV variance reduction is $1/(1 - \rho^2)$, which is 1097 against 482. Because $G \leq A$, the geometric option is more often worthless when the arithmetic one pays, and this mismatch matters more as K rises. (ii) $\beta$. The fitted $\beta$ is positive and above 1, rising from 1.029 to 1.068. Fixing $\beta = 1$ is therefore increasingly mis-specified out of the money: its VRF falls from 526 to 138, against 989 to 439 with pilot $\beta$. (iii) Antithetics help most in the money, where the payoff is nearly linear in Z (VRF 8.6 → 2.6), so their extra payoff cost pays off least out of the money. On point estimates, antithetic + pilot CV leads at the money (RE 614 vs 472), and pilot CV alone leads at $1.1S_0$ (387 vs 317). These two are statistically tied at every strike (overlapping CIs). Both dominate fixed $\beta = 1$ out of the money.

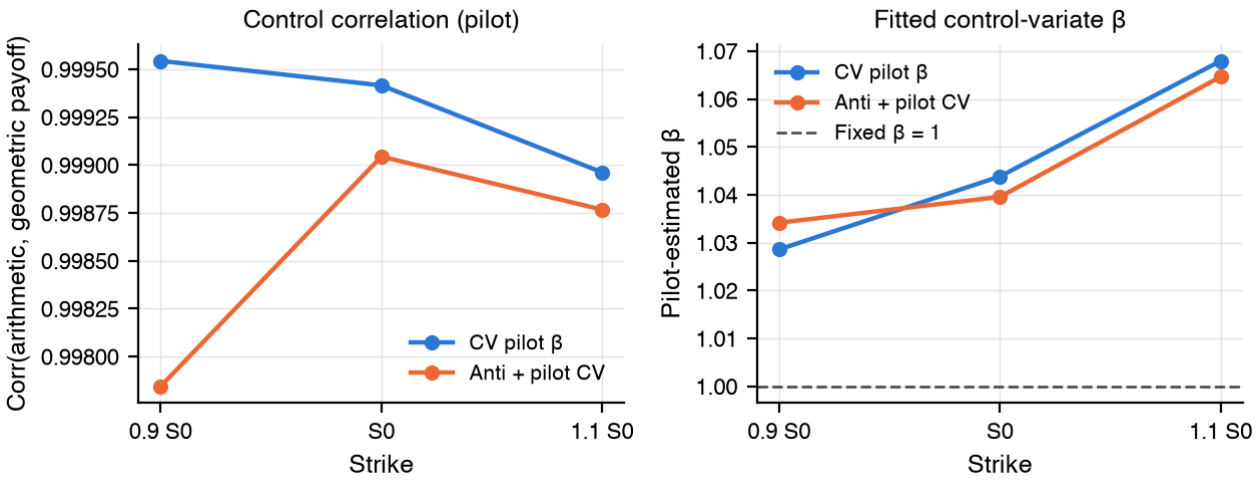



**Figure 14:** Pilot correlation between arithmetic and geometric payoffs (left) and fitted $\beta$ (right) by strike.

### 4.4 Production recommendation (Q3d)

**For routine use under a fixed time budget we recommend the antithetic + pilot-$\beta$ control-variate estimator.** At the money it has the lowest run-to-run SD of all five estimators (0.0039, against 0.1314 for crude) and the highest point RE (614). Its reported SE is accurate (SD/SE = 1.00), and estimating $\beta$ from an independent pilot keeps it unbiased. **We would reconsider** for deep out-of-the-money strikes, or whenever the pilot arithmetic–geometric correlation drops below about 0.999 and payoff evaluation dominates cost. There, at $1.1S_0$, the doubled payoff work of antithetic pairing is no longer repaid, and plain pilot-$\beta$ CV is the simpler, equally efficient choice.

## 5. Conclusion

Historical and implied volatility answer different questions. GARCH(1,1) is a marginally better P-forecaster, but every P forecast exceeds the risk-neutral variance implied by PLTR options, so only the implied 49.80% prices the benchmark call. Given that input, six independently implemented engines agree within 0.04 half-spreads, and each residual is traced to sampling noise or a quantified discretisation error. The one material gap, the risk-neutral GARCH scenario at +2.5 half-spreads, is a modelling assumption about the initial variance in an incomplete market. That is the model risk numerical agreement cannot reveal. For the Asian option, correct path dependence is essential, and pilot-$\beta$ geometric control variates, especially combined with antithetic pairs, are the efficient production choice.

## 6. Use of generative AI and other assistance

OpenAI ChatGPT, including Codex coding assistance, was used as a technical assistant to interpret the project brief; develop, debug and refactor Python and LaTeX code; propose validation tests; summarise executed outputs; improve exposition; and organise the submission files. It was not used as a source of market quotes: the stock and option observations come from the archived Yahoo Finance CSV files, and the reported tables and figures were generated by the submitted code with fixed random seeds.

Material outputs were checked against analytic Black–Scholes, put–call-parity and geometric-Asian benchmarks; convergence, stability, numerical-error and confidence-interval diagnostics; and deterministic reruns from the archived snapshot. The final code, modelling assumptions, economic interpretations and report wording were also reviewed manually. The group selected the modelling conventions and retains responsibility for every submitted claim. AI suggestions were included only after checking them against executable output or an identified source; no numerical result was accepted solely because an AI system asserted it. Each member separately completes the required individual GAI-assistance declaration.

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## Appendix A. Raw option-quote tables

Extract of the archived 9 October 2026 close snapshot, strikes 170–250 for every expiry (bid/ask in USD; mid = (bid + ask)/2). The complete chains (all strikes, calls and puts) are in Q1_Q2/Data/PLTR_calls_<expiry>.csv and PLTR_puts_<expiry>.csv.

**Table 18:** PLTR options expiring 2026-11-20 (42 days).

| K | Call bid | Call ask | Call mid | Vol. | OI | Put bid | Put ask | Put mid | Vol. | OI |
|---|---|---|---|---|---|---|---|---|---|---|
| 170.0 | 42.30 | 42.60 | 42.450 | 278 | 4911 | 2.53 | 2.64 | 2.585 | 1772 | 5556 |
| 175.0 | 38.10 | 38.35 | 38.225 | 65 | 4100 | 3.30 | 3.40 | 3.350 | 989 | 4327 |
| 180.0 | 34.10 | 34.35 | 34.225 | 351 | 8197 | 4.20 | 4.40 | 4.300 | 2300 | 8181 |
| 185.0 | 30.30 | 30.60 | 30.450 | 472 | 5523 | 5.50 | 5.60 | 5.550 | 592 | 7483 |
| 190.0 | 26.75 | 27.05 | 26.900 | 789 | 6445 | 6.90 | 7.05 | 6.975 | 1601 | 4587 |
| 195.0 | 23.50 | 24.25 | 23.875 | 702 | 4827 | 8.55 | 8.75 | 8.650 | 959 | 929 |
| 200.0 | 20.50 | 20.70 | 20.600 | 5180 | 12744 | 10.50 | 10.70 | 10.600 | 2470 | 2067 |
| 210.0 | 15.20 | 15.45 | 15.325 | 4857 | 7776 | 15.25 | 15.45 | 15.350 | 612 | 149 |
| 220.0 | 11.00 | 11.30 | 11.150 | 2706 | 8545 | 21.00 | 21.25 | 21.125 | 783 | 242 |
| 230.0 | 7.80 | 8.05 | 7.925 | 1327 | 7705 | 27.75 | 28.00 | 27.875 | 22 | 169 |
| 240.0 | 5.40 | 5.60 | 5.500 | 1409 | 3954 | 35.35 | 35.65 | 35.500 | 25 | 184 |
| 250.0 | 3.70 | 3.85 | 3.775 | 1876 | 7545 | 43.70 | 43.95 | 43.825 | 10 | 305 |

**Table 19:** PLTR options expiring 2026-12-18 (70 days).

| K | Call bid | Call ask | Call mid | Vol. | OI | Put bid | Put ask | Put mid | Vol. | OI |
|---|---|---|---|---|---|---|---|---|---|---|
| 170.0 | 44.65 | 45.00 | 44.825 | 202 | 9735 | 4.30 | 4.45 | 4.375 | 359 | 4368 |
| 175.0 | 40.70 | 41.20 | 40.950 | 201 | 52270 | 5.35 | 5.45 | 5.400 | 259 | 1914 |
| 180.0 | 36.95 | 37.25 | 37.100 | 308 | 4970 | 6.50 | 6.65 | 6.575 | 819 | 2851 |
| 185.0 | 33.40 | 33.70 | 33.550 | 151 | 14407 | 7.95 | 8.10 | 8.025 | 285 | 1460 |
| 190.0 | 30.05 | 30.95 | 30.500 | 367 | 4304 | 9.50 | 9.70 | 9.600 | 583 | 1390 |
| 195.0 | 26.90 | 27.75 | 27.325 | 200 | 2625 | 11.35 | 11.55 | 11.450 | 317 | 794 |
| 200.0 | 23.50 | 24.25 | 23.875 | 1357 | 7367 | 13.40 | 13.60 | 13.500 | 1634 | 1656 |
| 210.0 | 18.85 | 19.05 | 18.950 | 1691 | 4851 | 18.05 | 18.45 | 18.250 | 397 | 487 |
| 220.0 | 14.45 | 14.70 | 14.575 | 2486 | 7015 | 23.95 | 24.20 | 24.075 | 57 | 116 |
| 230.0 | 11.05 | 11.35 | 11.200 | 1327 | 3893 | 30.45 | 30.75 | 30.600 | 33 | 117 |
| 240.0 | 8.40 | 8.60 | 8.500 | 275 | 2935 | 37.75 | 38.10 | 37.925 | 4 | 34 |
| 250.0 | 6.30 | 6.40 | 6.350 | 1210 | 3299 | 45.60 | 46.05 | 45.825 | 2 | 65 |

**Table 20:** PLTR options expiring 2027-01-15 (98 days).

| K | Call bid | Call ask | Call mid | Vol. | OI | Put bid | Put ask | Put mid | Vol. | OI |
|---|---|---|---|---|---|---|---|---|---|---|
| 170.0 | 46.75 | 47.75 | 47.250 | 166 | 5319 | 5.80 | 6.20 | 6.000 | 102 | 5306 |
| 175.0 | 42.75 | 43.75 | 43.250 | 191 | 7228 | 7.00 | 7.45 | 7.225 | 399 | 3673 |
| 180.0 | 39.50 | 40.00 | 39.750 | 231 | 7789 | 8.40 | 8.75 | 8.575 | 124 | 6440 |
| 185.0 | 35.55 | 36.50 | 36.025 | 135 | 4551 | 9.85 | 10.30 | 10.075 | 307 | 2127 |
| 190.0 | 32.50 | 33.35 | 32.925 | 908 | 7129 | 11.75 | 12.15 | 11.950 | 189 | 3984 |
| 195.0 | 29.55 | 30.45 | 30.000 | 274 | 3755 | 13.40 | 14.00 | 13.700 | 198 | 2234 |
| 200.0 | 27.10 | 27.50 | 27.300 | 1213 | 18478 | 15.75 | 16.20 | 15.975 | 1645 | 2550 |
| 210.0 | 22.00 | 22.35 | 22.175 | 707 | 14296 | 20.60 | 21.25 | 20.925 | 414 | 733 |
| 220.0 | 17.75 | 18.20 | 17.975 | 597 | 8959 | 26.20 | 27.00 | 26.600 | 28 | 571 |
| 230.0 | 14.10 | 14.50 | 14.300 | 2194 | 5695 | 32.60 | 34.30 | 33.450 | 2 | 159 |
| 240.0 | 11.10 | 11.60 | 11.350 | 401 | 7235 | 39.10 | 41.45 | 40.275 | 1 | 137 |
| 250.0 | 8.85 | 9.40 | 9.125 | 1551 | 12329 | 46.90 | 49.05 | 47.975 | 27 | 169 |

**Table 21:** PLTR options expiring 2027-02-19 (133 days).

| K | Call bid | Call ask | Call mid | Vol. | OI | Put bid | Put ask | Put mid | Vol. | OI |
|---|---|---|---|---|---|---|---|---|---|---|
| 170.0 | 50.00 | 51.75 | 50.875 | 225 | 808 | 8.95 | 9.45 | 9.200 | 97 | 932 |
| 175.0 | 46.55 | 48.70 | 47.625 | 92 | 569 | 10.45 | 10.95 | 10.700 | 30 | 517 |
| 180.0 | 43.35 | 44.80 | 44.075 | 239 | 790 | 12.05 | 12.55 | 12.300 | 33 | 431 |
| 185.0 | 40.15 | 41.45 | 40.800 | 12 | 1387 | 13.85 | 14.35 | 14.100 | 30 | 347 |
| 190.0 | 37.60 | 38.80 | 38.200 | 50 | 2424 | 15.80 | 16.30 | 16.050 | 71 | 429 |
| 195.0 | 34.90 | 36.05 | 35.475 | 69 | 656 | 17.85 | 18.45 | 18.150 | 23 | 526 |
| 200.0 | 32.20 | 32.95 | 32.575 | 340 | 3434 | 20.15 | 20.70 | 20.425 | 93 | 372 |
| 210.0 | 27.30 | 28.10 | 27.700 | 686 | 3247 | 25.05 | 25.50 | 25.275 | 39 | 144 |
| 220.0 | 23.00 | 23.80 | 23.400 | 202 | 1066 | 30.75 | 31.75 | 31.250 | 11 | 78 |
| 230.0 | 19.50 | 20.10 | 19.800 | 973 | 1265 | 36.60 | 38.60 | 37.600 | 9 | 84 |
| 240.0 | 16.25 | 16.90 | 16.575 | 184 | 1191 | 43.10 | 45.40 | 44.250 | 1 | 113 |
| 250.0 | 13.70 | 14.10 | 13.900 | 346 | 905 | 50.60 | 52.85 | 51.725 | 3 | 169 |

## Appendix B. Code appendix and reproducibility record

**Reproduction.** Python $\geq$ 3.12. From the project root: create a virtual environment, run pip install -r requirements.txt, then python run_all.py (about 1 minute on an Apple M5; works on Windows, macOS and Linux). The script executes notebooks 01–10 top to bottom on the archived CSVs (no downloads), runs the Q3 validation tests and simulation, writes every table, figure and results/results_manifest.json, and regenerates this report with report/build_final_report.py. The full environment used for the reported run is in requirements-lock.txt.

**Table 22:** Code map.

| Question | File | Content (all pricing code self-implemented) |
|---|---|---|
| Q1a | Notebooks/01_data_audit.ipynb, 02_option_audit.ipynb | Price and option-chain audit, split checks, calibration/validation split |
| Q1b | Notebooks/03_Historical_volatility.ipynb | GBM, EWMA ($\lambda$ selection), GARCH(1,1) via arch MLE, validation MSE, shock responses |
| Q1c–d | Notebooks/04_Implied_volatility.ipynb | BS pricer, Brent IV inversion, smiles, term structure, forward variance, P vs Q, parity spot |
| Q2a | Notebooks/05_BS_benchmark.ipynb | BS price and Greeks, put–call parity |
| Q2b | Notebooks/06_Monte_Carlo.ipynb | Exact-GBM MC price, CRN central-difference and pathwise delta, bump experiment |
| Q2c | Notebooks/07_Binomial_Trees.ipynb | CRR and JR trees, Greeks, admissibility, oscillation diagnostic |
| Q2d | Notebooks/08_Finite_Difference.ipynb | Heat-equation explicit/implicit/CN, Thomas LU, stability, truncation, refinement |
| Q2e | Notebooks/09_risk_neutral_GARCH_scenario.ipynb | GARCH-Q simulation, GBM control variate, $h_0$ sensitivity, martingale check |
| Q2f | Notebooks/10_Reconciliation_Model_Risk.ipynb | Reconciliation table and discrepancy checks |
| Q3 | Q3/FN6808_Q3/src/q3_simulation.py (+ config/config.json) | Validation tests, path generators, five estimators, budget accounting, 30-seed study, bootstrap RE |
| All | Q1_Q2/Src/fn6808_common.py, tools/finalize_results.py, run_all.py | Paths, seed registry, figure style, manifest writer, pipeline |

**Table 23:** Random seeds.

| Experiment | Seed |
|---|---|
| Q2 MC price (N = $2^{12}\ldots2^{18}$) | SeedSequence(20261009).spawn(17)[0–3] $\rightarrow$ PCG64 |
| Q2 MC delta / bump experiment | children [4] / [5] |
| Q2e GARCH-Q convergence (1k…200k paths) | children [6–11] |
| Q2e $h_0$ sensitivity (4 cases) / CV pilot | children [12–15] / [16] |
| Q3 macro-repetitions | master 20261019; independent SeedSequence streams (repetition, estimator, stream) |
| Q3 part (a) / bootstrap | SeedSequence([20261019, 1000]) / SeedSequence([20261019, 2000]) |

**Table 24:** Software and hardware.

| Item | Version |
|---|---|
| Python | 3.13.9 |
| NumPy | 2.5.4 |
| pandas | 3.0.6 |
| SciPy | 1.18.1 |
| Matplotlib | 3.11.2 |
| arch (GARCH MLE only) | 8.0.0 |
| Machine | Apple M5, 10 cores, 16 GB, macOS-26.5.2-arm64-arm-64bit-Mach-O |
| Lock file | requirements-lock.txt |